

Project Brief

Arabic Social Media Analytics — Power BI

1. Project Overview

In this capstone project, you will work with a large, realistic dataset of Arabic-language social media posts. The raw file is intentionally messy — the way real exported data usually looks. Your task is to clean it, build a proper star-schema data model in Power BI, write DAX measures, and design a dashboard that answers a set of business analysis questions you choose.

This project is meant to mirror a real analytics workflow end-to-end: Raw Data → Cleaning → Data Modeling → DAX → Dashboard → Insights.

2. Dataset

File provided: Social_Media_Raw_Arabic.xlsx (also available as .csv). The dataset contains 4,300+ rows of social media post performance data in Arabic, covering multiple pages/accounts, platforms, and content categories over 2024–2025.

2.1 Columns in the raw file

Column	Description
Post_ID	Unique identifier for each post (a few are missing — data entry gaps)
Account_Name	Page/account name in Arabic (inconsistent spacing, some duplicated with symbols)
Platform	Facebook / Instagram / TikTok / Twitter(X) / YouTube — written inconsistently (Arabic, English, abbreviations)
Category	Content category in Arabic (Cooking, Sports, Fashion, Tech, News, Entertainment, Education) — inconsistent labels
Post_Type	Video / Image / Text / Reel / Story — inconsistent labels (Arabic/English mixed)
Post_Date	Post date in mixed formats (YYYY-MM-DD, DD/MM/YYYY, DD-MM-YYYY), a few blank
Post_Text	Short Arabic post caption/title (some with extra whitespace)
Followers_At_Post	Account follower count at time of posting
Reach	Number of unique accounts reached (some missing)
Impressions	Total times the post was displayed
Likes	Number of likes (some negative errors, some as text with commas, some missing)
Comments_Count	Number of comments (a few extreme outliers)
Shares	Number of shares (some missing)
Is_Sponsored	Whether the post was a paid/sponsored post — inconsistent Yes/No/1/0/لا/نعم, mostly missing

Note: This dataset is synthetic (generated for training purposes) but designed to reflect realistic patterns and realistic data-quality problems.

3. Requirements

3.1 Data Cleaning (Power Query)

Clean the raw table in Power Query before loading it into the model. At minimum, handle:

- Remove exact duplicate rows
- Standardize Platform, Category, and Post_Type values into single consistent categories (merge all Arabic/English/abbreviated variants)
- Trim extra spaces from text columns (Account_Name, Post_Text, etc.)
- Parse Post_Date into a proper Date type despite mixed formats; decide how to handle blank dates
- Convert Likes to a proper numeric type (remove thousand separators stored as text)
- Fix or flag negative values in Likes (data-entry errors)
- Decide how to treat extreme outliers in Comments_Count (investigate, cap, or flag — document your reasoning)
- Handle missing values
- Standardize Is_Sponsored into a clean Yes/No (or True/False) field; decide how to treat the large number of missing values
- Remove rows with a missing

3.2 Data Modeling — Star Schema

After cleaning, build a star schema. Do not keep one flat table as your model. Required tables:

Table	Type	Notes
Fact_Posts	Fact table	One row per post: keys to all dimensions + Likes, Comments_Count, Shares, Reach, Impressions, Followers_At_Post
Dim_Date	Dimension	Built from Post_Date: Date, Day, Month, MonthName, Quarter, Year, DayOfWeek, IsWeekend (use a full calendar table, marked as a Date Table in Power BI)
Dim_Account	Dimension	Account_ID (surrogate key), Account_Name, Category (one row per account)
Dim_Platform	Dimension	Platform_ID, Platform_Name (cleaned, one row per platform)
Dim_PostType	Dimension	PostType_ID, PostType_Name (cleaned, one row per post type)

Set up relationships as one-to-many from each dimension to Fact_Posts, all filtering in a single direction. Mark Dim_Date as an official Date Table.

3.3 DAX Measures (required)

Create at least the following measures — add more if needed to answer your chosen questions:

- Total Likes, Total Comments, Total Shares, Total Reach, Total Impressions
- Engagement Rate
- Average Engagement Rate per Post

- MOM Growth % for Total Likes (month-over-month)
- YOY Growth % for Total Engagement
- Posts Count (distinct count of Post_ID)
- Top Category by Engagement (using RANKX or a ranking measure)
- % of Sponsored vs Organic Engagement

3.4 Dashboard Requirements

- At least 2 dashboard pages
- A mix of visuals: KPI cards, a trend line/area chart over time, a bar/column chart comparing categories or platforms, and at least one matrix or table
- At least one slicer (e.g., Date, Platform, or Category) that filters the whole page
- Consistent color scheme and clear titles on every visual
- Tooltips or data labels used meaningfully, not just for decoration

4. Analysis Questions Bank

Below are 16 possible analysis questions. Choose 4–5 questions to design your dashboard around. Your chosen visuals and DAX measures should directly answer the questions you pick.

1. Which platform generates the highest average engagement rate?
2. Which content category performs best in terms of Likes and Comments?
3. How does engagement change month over month across 2024–2025?
4. Which day of the week gets the highest average Reach?
5. Is there a meaningful difference in engagement between Video, Image, Reel, Story, and Text posts?
6. Which accounts/pages are the top 5 performers by total engagement?
7. Do sponsored posts get significantly more Reach or Engagement than organic posts?
8. How does Impressions-to-Reach ratio vary by platform?
9. Which category has grown or declined the most year-over-year?
10. Is there a relationship between an account's follower count and its engagement rate?
11. What is the best-performing combination of Platform + Post_Type?
12. How consistent is posting activity across months (which months have the most/fewest posts)?
13. Which category relies most heavily on sponsored content?
14. What share of total engagement comes from the top 20% of posts (a simple concentration/Pareto view)?
15. How does average Comments_Count differ between weekday and weekend posts?
16. Which platform has the most stable (least volatile) engagement rate over time?

5. Deliverables

- Cleaned Power BI file (.pbix) with the star schema model, relationships, and all DAX measures
- Dashboard (within the same .pbix) covering your chosen 4–5 questions
- A short written summary (half a page to one page) listing: the questions you chose, your key findings, and one actionable recommendation per question

6. Submission Checklist

- .pbix file opens without errors and refreshes cleanly
- Star schema is visible in Model view (no single flat table)
- Dim_Date is marked as a Date Table
- All required DAX measures exist and are named clearly
- Dashboard has at least 2 pages with a working slicer
- Written summary of insights is attached
- File is named: FirstName_LastName_SocialMediaCapstone.pbix